

33.3 Repeated measures

A repeated-measures study observes the same subject on several occasions: a clustered design whose cluster is the subject and whose “treatments” are the occasions, laid out like the split-plot design of [Section 33.1](#), and for many years analysed that way.

The analogy has one weak joint. What makes the split-plot covariance compound symmetric is the *randomization* of subplot treatments within the whole plot, and occasions cannot be randomized: measurements close together in time are more alike than distant ones, so compound symmetry here is usually false. What is remarkable is that the univariate F test survives a much weaker condition, and that the condition can be written down exactly.

33.3.1 The univariate analysis

Let $\mathbf{y}_1, \dots, \mathbf{y}_N$ be independent $N_m(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ vectors, one per subject, with $\boldsymbol{\Sigma}$ positive definite and unknown, and write y_{ij} for occasion j of subject i . The hypothesis of interest is

$$H : \mu_1 = \dots = \mu_m, \quad \text{that is,} \quad \boldsymbol{\mu} \in \text{span}\{\mathbf{1}\}.$$

Laid out with subjects as rows and occasions as columns the data have one observation per cell, and the classical analysis is the additive two-way analysis of variance of [Section 16.1](#), with the subject-by-occasion interaction as error. With $\bar{y}_{i\cdot}$, $\bar{y}_{\cdot j}$ and $\bar{y}$ the subject, occasion and grand means,

$$SS_T = N \sum_{j=1}^m (\bar{y}_{\cdot j} - \bar{y})^2, \quad SS_E = \sum_{i=1}^N \sum_{j=1}^m (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y})^2, \quad (33.3.1)$$

and the statistic is $F = \{SS_T/(m-1)\}/\{SS_E/((N-1)(m-1))\}$. It is the subplot stratum of a split-plot design with one whole plot per subject, and under compound symmetry it is exact ([Theorem 33.3.2\(b\)](#) below, with [Exercise \(B1\)](#)). The question is how much of that exactness survives a general $\boldsymbol{\Sigma}$.

Everything reduces to the within-subject contrasts. Let $\mathbf{U}$ be $m \times (m-1)$ with orthonormal columns orthogonal to $\mathbf{1}$, so that $\mathbf{U}\mathbf{U}^\top = \mathbf{I} - \bar{\mathbf{J}}_m$, and set $\mathbf{z}_i = \mathbf{U}^\top \mathbf{y}_i$: independent $N_{m-1}(\mathbf{U}^\top \boldsymbol{\mu}, \boldsymbol{\Psi})$ with $\boldsymbol{\Psi} = \mathbf{U}^\top \boldsymbol{\Sigma} \mathbf{U}$, with H saying exactly that $\mathbf{U}^\top \boldsymbol{\mu} = \mathbf{0}$. The subject means carry no information about H .

33.3.2 Sphericity

Definition 33.3.1 — Sphericity and Box's measure

Let $\boldsymbol{\Sigma}$ be an $m \times m$ positive definite matrix and $\mathbf{U}$ as above. $\boldsymbol{\Sigma}$ satisfies **sphericity** if $\mathbf{U}^\top \boldsymbol{\Sigma} \mathbf{U} = \lambda \mathbf{I}_{m-1}$ for some $\lambda > 0$. **Box's measure of sphericity** is

$$\epsilon(\boldsymbol{\Sigma}) = \frac{\{\text{tr } \boldsymbol{\Psi}\}^2}{(m-1) \text{tr}(\boldsymbol{\Psi}^2)}, \quad \boldsymbol{\Psi} = \mathbf{U}^\top \boldsymbol{\Sigma} \mathbf{U}. \quad (33.3.2)$$

Neither the condition nor ϵ depends on the choice of $\mathbf{U}$: two such matrices satisfy $\tilde{\mathbf{U}} = \mathbf{U}\mathbf{Q}$ with $\mathbf{Q}$ orthogonal, so $\tilde{\Psi} = \mathbf{Q}^\top \Psi \mathbf{Q}$ has the same trace, the same trace of square, and is a multiple of the identity exactly when Ψ is.

Lemma 33.3.1 — Equivalent forms of sphericity

For a positive definite $m \times m$ matrix Σ the following are equivalent:

- (i) $\mathbf{U}^\top \Sigma \mathbf{U} = \lambda \mathbf{I}_{m-1}$;
- (ii) $\text{Var}(Y_j - Y_k) = 2\lambda$ for every $j \neq k$;
- (iii) $\sigma_{jk} = a_j + a_k + \lambda \delta_{jk}$ for some $a_1, \dots, a_m$, where δ_{jk} is one if $j = k$ and zero otherwise;
- (iv) $\Sigma = \lambda \mathbf{I} + \mathbf{1}\mathbf{a}^\top + \mathbf{a}\mathbf{1}^\top$ for some $\mathbf{a} \in \mathbb{R}^m$.

Moreover $1/(m-1) \leq \epsilon(\Sigma) \leq 1$, with $\epsilon = 1$ iff sphericity holds, and compound symmetry implies sphericity but not conversely.

Proof. Write $\mathbf{C} = \mathbf{I} - \bar{\mathbf{J}}_m = \mathbf{U}\mathbf{U}^\top$. Since $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_{m-1}$, condition (i) holds iff $\mathbf{C}\Sigma\mathbf{C} = \lambda\mathbf{C}$.

- (i) $\Rightarrow$ (ii): $\mathbf{e}_j - \mathbf{e}_k$ is orthogonal to $\mathbf{1}$, hence fixed by $\mathbf{C}$, so $\text{Var}(Y_j - Y_k)$ equals $(\mathbf{e}_j - \mathbf{e}_k)^\top \mathbf{C}\Sigma\mathbf{C}(\mathbf{e}_j - \mathbf{e}_k)$, which is $\lambda\|\mathbf{e}_j - \mathbf{e}_k\|^2 = 2\lambda$.
- (ii) $\Rightarrow$ (iii): $\text{Var}(Y_j - Y_k) = \sigma_{jj} + \sigma_{kk} - 2\sigma_{jk} = 2\lambda$ gives $\sigma_{jk} = (\sigma_{jj} - \lambda)/2 + (\sigma_{kk} - \lambda)/2$ for $j \neq k$. Put $a_j = (\sigma_{jj} - \lambda)/2$; then $\sigma_{jk} = a_j + a_k$ off the diagonal and $\sigma_{jj} = 2a_j + \lambda$ on it, which is (iii).
- (iii) $\Leftrightarrow$ (iv) is the same statement in matrix form.
- (iv) $\Rightarrow$ (i): $\mathbf{C}\mathbf{1} = \mathbf{0}$, so $\mathbf{C}\Sigma\mathbf{C} = \lambda\mathbf{C}$.

For the bounds, let $\psi_1, \dots, \psi_{m-1} > 0$ be the eigenvalues of Ψ . Then $\epsilon = (\sum_l \psi_l)^2 / \{(m-1) \sum_l \psi_l^2\}$. The Cauchy–Schwarz inequality gives $(\sum_l \psi_l)^2 \leq (m-1) \sum_l \psi_l^2$ with equality iff all ψ_l are equal, so $\epsilon \leq 1$ with equality iff $\Psi = \lambda \mathbf{I}$; and $\sum_l \psi_l^2 \leq (\sum_l \psi_l)^2$ for nonnegative numbers gives $\epsilon \geq 1/(m-1)$. Compound symmetry is (iv) with all a_j equal; taking the a_j unequal gives sphericity without compound symmetry, and such a Σ is positive definite as soon as λ is large enough. ■

Condition (ii) is the form to remember and to check: *all pairwise differences of the repeated measurements have the same variance*. Compound symmetry says more than the F test needs.

Theorem 33.3.2 — Sphericity is necessary and sufficient for the exact F test

Let $\mathbf{y}_1, \dots, \mathbf{y}_N$ be independent $N_m(\boldsymbol{\mu}, \Sigma)$ with Σ positive definite, $m \geq 2$, $N \geq 2$, and let F be the statistic built from Eq. 33.3.1. Put $\Psi = \mathbf{U}^\top \Sigma \mathbf{U}$, $\mathbf{q} = \sqrt{N} \bar{\mathbf{z}}$ and $\mathbf{W} = \sum_i (\mathbf{z}_i - \bar{\mathbf{z}})(\mathbf{z}_i - \bar{\mathbf{z}})^\top$.

- (a) $\text{SS}_T = \|\mathbf{q}\|^2$ and $\text{SS}_E = \text{tr}(\mathbf{W})$, and under H the vector $\mathbf{q} \sim N_{m-1}(\mathbf{0}, \Psi)$ is

independent of $\mathbf{W} \sim W_{m-1}(N-1, \mathbf{\Psi})$. Hence $F = N(N-1)\|\bar{\mathbf{z}}\|^2 / \text{tr}(\mathbf{W})$.

- (b) If $\mathbf{\Sigma}$ satisfies sphericity, then under H $F \sim F(m-1, (N-1)(m-1))$ exactly, for every $N \geq 2$.
- (c) Conversely, if the null distribution of F is $F(m-1, (N-1)(m-1))$ for every $N \geq 2$, then $\mathbf{\Sigma}$ satisfies sphericity.

Proof.

- (a) Since $\mathbf{C} = \mathbf{U}\mathbf{U}^\top$ and $\mathbf{U}$ has orthonormal columns, $\sum_j (\bar{y}_{\cdot j} - \bar{y})^2 = \|\mathbf{C}\bar{\mathbf{y}}\|^2 = \|\bar{\mathbf{z}}\|^2$, so $\text{SS}_T = N\|\bar{\mathbf{z}}\|^2 = \|\mathbf{q}\|^2$; likewise subject i 's contribution to SS_E is $\|\mathbf{z}_i - \bar{\mathbf{z}}\|^2$, summing to $\text{tr}(\mathbf{W})$. Under H the $\mathbf{z}_i$ are independent $N_{m-1}(\mathbf{0}, \mathbf{\Psi})$, so Lemma 14.2.3 gives the Wishart law of $\mathbf{W}$ and its independence of $\bar{\mathbf{z}}$. Finally $F = \{\|\mathbf{q}\|^2 / (m-1)\} \{(N-1)(m-1) / \text{tr} \mathbf{W}\}$.
- (b) If $\mathbf{\Psi} = \lambda \mathbf{I}_{m-1}$, then $\mathbf{q}/\sqrt{\lambda}$ is standard normal, so $\|\mathbf{q}\|^2/\lambda \sim \chi^2(m-1)$. And $\mathbf{W}/\lambda \sim W_{m-1}(N-1, \mathbf{I})$, which is the matrix of sums of squares and cross products of $(N-1)(m-1)$ independent standard normal variables, so $\text{tr}(\mathbf{W})/\lambda \sim \chi^2((N-1)(m-1))$. The two are independent, and λ cancels from the ratio, leaving the ratio of two independent chi-squared variables each divided by its degrees of freedom (Definition 4.1.2).
- (c) Write F_N for the statistic with N subjects. By (a),

$$F_N = \|\mathbf{q}\|^2 \cdot \frac{N-1}{\text{tr}(\mathbf{W})},$$

where $\mathbf{q} \sim N_{m-1}(\mathbf{0}, \mathbf{\Psi})$ has the *same* distribution for every N and is independent of $\mathbf{W}$. Since $\mathbf{W}/(N-1)$ is an average of $N-1$ independent $W_{m-1}(1, \mathbf{\Psi})$ matrices, the weak law of large numbers gives $\text{tr}(\mathbf{W})/(N-1) \rightarrow \text{tr} \mathbf{\Psi}$ in probability, so by Slutsky's lemma $F_N \rightarrow \|\mathbf{q}\|^2 / \text{tr} \mathbf{\Psi}$ in distribution. On the other hand $F(m-1, \nu) \rightarrow \chi^2(m-1)/(m-1)$ in distribution as $\nu \rightarrow \infty$, because its denominator chi-squared divided by ν tends to one in probability. If the exactness holds for every N , the two limits agree:

$$\|\mathbf{q}\|^2 \stackrel{d}{=} \frac{\text{tr} \mathbf{\Psi}}{m-1} \chi^2(m-1).$$

Now match variances. By Theorem 4.2.3, $\text{Var}(\|\mathbf{q}\|^2) = 2 \text{tr}(\mathbf{\Psi}^2)$, while the right-hand side has variance $2\{\text{tr} \mathbf{\Psi}\}^2/(m-1)$. Equality says $(m-1) \text{tr}(\mathbf{\Psi}^2) = \{\text{tr} \mathbf{\Psi}\}^2$, that is $\epsilon(\mathbf{\Sigma}) = 1$, and Lemma 33.3.1 makes that sphericity. ■

Idea. The repeated-measures F test does not require the occasions to be exchangeable, only that the $m-1$ orthonormalized within-subject contrasts be uncorrelated with equal variances — that the contrast space look spherical, which series of measurements in time almost never do.

33.3.3 Two corrections and one alternative

When sphericity fails, $\|\mathbf{q}\|^2$ and $\text{tr}(\mathbf{W})$ are still weighted sums of chi-squared variables, and matching their first two moments to scaled chi-squared variables gives a usable approximation (Box 1954).

Proposition 33.3.3 — The moment-matched approximation

Under the assumptions of [Theorem 33.3.2](#), with H true, write $\epsilon = \epsilon(\mathbf{\Sigma})$, $d_1 = m - 1$ and $d_2 = (N - 1)(m - 1)$. Then

- (a) $E SS_T = \text{tr } \mathbf{\Psi}$, $\text{Var } SS_T = 2 \text{tr}(\mathbf{\Psi}^2)$, $E SS_E = (N - 1) \text{tr } \mathbf{\Psi}$ and $\text{Var } SS_E = 2(N - 1) \text{tr}(\mathbf{\Psi}^2)$.
- (b) The scaled chi-squared variable $g\chi^2(h)$ matching the first two moments of SS_T has $h = \epsilon d_1$ and $g = \text{tr}(\mathbf{\Psi}^2) / \text{tr } \mathbf{\Psi}$; the one matching SS_E has $h' = \epsilon d_2$ and the *same* g . Since SS_T and SS_E are independent by [Theorem 33.3.2\(a\)](#), replacing each by its matched scaled chi-squared gives the approximation $F \approx F(\epsilon d_1, \epsilon d_2)$, which is exact when $\epsilon = 1$.

Proof.

- (a) The first two are [Theorem 4.2.3](#) with $\mathbf{A} = \mathbf{I}$ and mean zero. Writing $\mathbf{W} = \sum_{l=1}^{N-1} \mathbf{w}_l \mathbf{w}_l^\top$ with $\mathbf{w}_l$ independent $N_{m-1}(\mathbf{0}, \mathbf{\Psi})$ ([Lemma 14.2.3](#)), $\text{tr } \mathbf{W} = \sum_l \|\mathbf{w}_l\|^2$ is a sum of $N - 1$ independent copies of $\|\mathbf{q}\|^2$, so its mean and variance are $N - 1$ times theirs.
- (b) Matching $gh = \text{tr } \mathbf{\Psi}$ and $2g^2h = 2 \text{tr}(\mathbf{\Psi}^2)$ gives $g = \text{tr}(\mathbf{\Psi}^2) / \text{tr } \mathbf{\Psi}$ and $h = \{\text{tr } \mathbf{\Psi}\}^2 / \text{tr}(\mathbf{\Psi}^2) = \epsilon d_1$. For SS_E both moments are multiplied by $N - 1$, so g is unchanged and $h' = (N - 1)h = \epsilon d_2$. Since $d_2/d_1 = h'/h$, the ratio $(SS_T/d_1)/(SS_E/d_2)$ becomes $\{\chi^2(h)/h\}/\{\chi^2(h')/h'\}$, an $F(h, h')$ variable, the two chi-squared variables being independent by [Theorem 33.3.2\(a\)](#); at $\epsilon = 1$ this is the exact result of [Theorem 33.3.2\(b\)](#). ■

By [Lemma 33.3.1](#), $\epsilon \geq 1/(m - 1)$, so the approximating distribution is never more dispersed than $F(d_1/(m - 1), d_2/(m - 1)) = F(1, N - 1)$, the largest that [Proposition 33.3.3\(b\)](#) can ask for. Referring F to $F(1, N - 1)$ is therefore conservative within Box's approximation, and Geisser and Greenhouse (1958) proved it exactly conservative, for every $\mathbf{\Sigma}$ and N . This is the **lower-bound** correction, turned into a recommendation by Greenhouse and Geisser (1959).

Part (b) becomes a procedure once ϵ is estimated from the sample covariance matrix $\mathbf{S}$ of the $\mathbf{y}_i$.

- The **Greenhouse–Geisser** correction refers F to $F(\hat{\epsilon}d_1, \hat{\epsilon}d_2)$ with the plug-in $\hat{\epsilon} = \epsilon(\mathbf{S})$, which is biased downward, so the test is conservative.
- The **Huynh–Feldt** correction removes the leading bias. For a single group of N subjects it uses

$$\tilde{\epsilon} = \min\left\{1, \frac{N(m - 1)\hat{\epsilon} - 2}{(m - 1)\{(N - 1) - (m - 1)\hat{\epsilon}\}}\right\}, \quad (33.3.3)$$

which exceeds $\hat{\epsilon}$ and gives a less conservative test; its own bias is upward when ϵ is small.

- The **lower-bound** correction takes $\epsilon = 1/(m - 1)$: safe, and usually too conservative to be useful.

A third option avoids the issue. Since the $\mathbf{z}_i$ are independent $N_{m-1}(\mathbf{U}^\top \boldsymbol{\mu}, \boldsymbol{\Psi})$ with $\boldsymbol{\Psi}$ unrestricted, H says a multivariate normal mean is zero, and the classical test is Hotelling's.

Proposition 33.3.4 — The multivariate test

Under the assumptions of Theorem 33.3.2 with $N \geq m$, let $\mathbf{S}_z = \mathbf{W}/(N - 1)$ and

$$T^2 = N \bar{\mathbf{z}}^\top \mathbf{S}_z^{-1} \bar{\mathbf{z}}.$$

Then T^2 does not depend on the choice of $\mathbf{U}$, nor on replacing $\mathbf{U}$ by $\mathbf{U}\mathbf{K}$ for any nonsingular $(m - 1) \times (m - 1)$ matrix $\mathbf{K}$; and under H ,

$$\frac{N - m + 1}{(N - 1)(m - 1)} T^2 \sim F(m - 1, N - m + 1),$$

whatever $\boldsymbol{\Sigma}$ is. The distributional statement is Hotelling's and is not proved here; see Anderson (2003, §5.2).

Proof. Only the invariance is proved. Replacing $\mathbf{U}$ by $\mathbf{U}\mathbf{K}$ replaces $\bar{\mathbf{z}}$ by $\mathbf{K}^\top \bar{\mathbf{z}}$ and $\mathbf{S}_z$ by $\mathbf{K}^\top \mathbf{S}_z \mathbf{K}$, leaving T^2 unchanged; two orthonormal bases differ by an orthogonal $\mathbf{K}$. The requirement $N \geq m$ makes $\mathbf{S}_z$, a Wishart matrix on $N - 1 \geq m - 1$ degrees of freedom, invertible with probability one. ■

Exactness under any $\boldsymbol{\Sigma}$ sounds decisive, but it costs power: the multivariate test estimates a whole covariance matrix and has $N - m + 1$ denominator degrees of freedom rather than $(N - 1)(m - 1)$. When sphericity nearly holds the univariate test is much more powerful, when it fails badly the two are comparable, and when $N < m$ the multivariate test does not exist.

33.3.4 An example, four ways

Example. (Grip strength over six visits)

Twenty-four patients have grip strength measured, in kilograms, at six weekly visits, simulated from $N_6(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ first-order autoregressive, $\sigma_{jk} = 4 \times 0.6^{|j-k|}$, and a mean drifting slowly upward; the observed occasion means are 47.818, 48.099, 49.056, 49.177, 49.561 and 49.263 kg. The true Box measure is $\epsilon = 0.700$: adjacent visits correlate 0.6 and visits five weeks apart only 0.6^5 , so adjacent differences are much less variable than distant ones. The four analyses of H give

Analysis	statistic	degrees of freedom	p
univariate, uncorrected	4.770	5, 115	0.0005

Greenhouse– Geisser, $\hat{\epsilon} = 0.645$	4.770	3.23, 74.2	0.0035
Huynh–Feldt, $\tilde{\epsilon} = 0.763$	4.770	3.81, 87.7	0.0019
multivariate	3.605	5, 19	0.0184

All four reject, but the evidence differs by a factor of nearly forty between first and last. Distrust the uncorrected test: its nominal degrees of freedom are too large by the factor ϵ .

```
import numpy as np
from scipy import stats

N, m = 24, 6
mu_true = np.array([48.0, 48.5, 48.9, 49.2, 49.4, 49.5]) # kg, a slow improvement
sigma, rho_true = 2.0, 0.6
lags = np.abs(np.subtract.outer(np.arange(m), np.arange(m)))
Sigma_true = sigma**2 * rho_true**lags # AR(1) within a patient
rng = np.random.default_rng(3303)
Yobs = np.round(mu_true + rng.multivariate_normal(np.zeros(m), Sigma_true, N), 2)

def contrast_basis(m):
    """Orthonormal basis U of the space of contrasts among m repeated measures."""
    Q, _ = np.linalg.qr(np.column_stack([np.ones(m), np.eye(m)[1:]]))
    return Q[:, 1:] # columns orthonormal and
                    # orthogonal to 1

def epsilon(Psi):
    """Box's measure of departure from sphericity, 1/(m-1) <= eps <= 1."""
    k = Psi.shape[0]
    return np.trace(Psi)**2 / (k * np.sum(Psi * Psi))

U = contrast_basis(m)
Z = Yobs @ U # N x (m-1) contrasts, one row
              # per patient
zbar = Z.mean(axis=0)
Sz = np.cov(Z, rowvar=False, ddof=1) # Wishart, (N-1) degrees of
                                     # freedom

F_uncorrected = N * (zbar @ zbar) / np.trace(Sz)
eps_hat = epsilon(Sz)
eps_hf = min(1.0, (N * (m - 1) * eps_hat - 2)
              / ((m - 1) * ((N - 1) - (m - 1) * eps_hat)))
d1, d2 = m - 1, (N - 1) * (m - 1)
p_un = stats.f.sf(F_uncorrected, d1, d2)
p_gg = stats.f.sf(F_uncorrected, d1 * eps_hat, d2 * eps_hat)
p_hf = stats.f.sf(F_uncorrected, d1 * eps_hf, d2 * eps_hf)
T2 = N * zbar @ np.linalg.solve(Sz, zbar) # Hotelling's statistic
F_mv = (N - m + 1) / ((N - 1) * (m - 1)) * T2
p_mv = stats.f.sf(F_mv, m - 1, N - m + 1)
print(f"F = {F_uncorrected:.3f} on ({d1}, {d2}) df, p = {p_un:.4f}")
print(f"Greenhouse-Geisser eps = {eps_hat:.3f}, p = {p_gg:.4f}")
print(f"Huynh-Feldt eps = {eps_hf:.3f}, p = {p_hf:.4f}")
print(f"multivariate F({m - 1}, {N - m + 1}) = {F_mv:.3f}, p = {p_mv:.4f}")
```

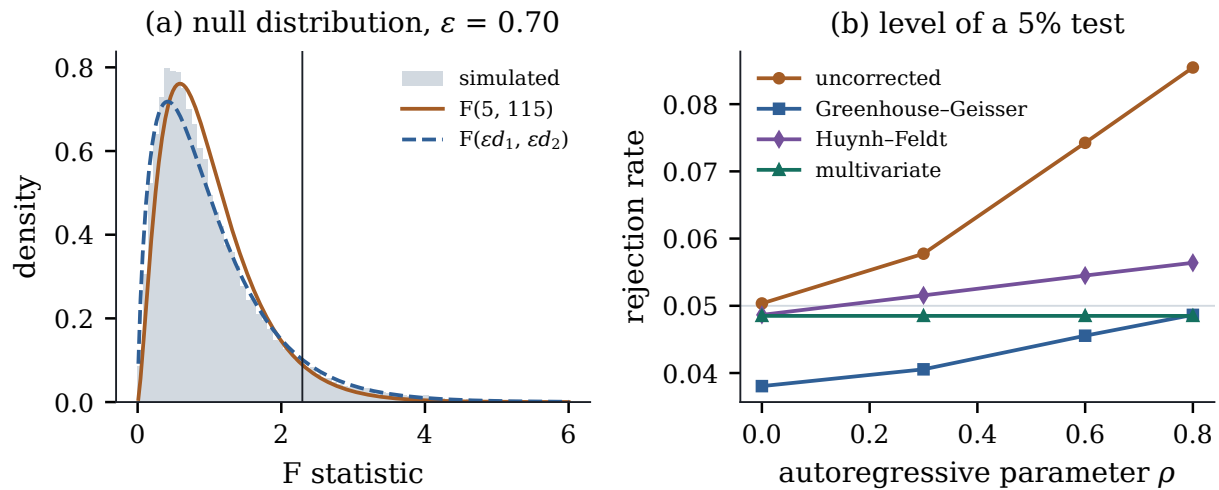


Figure 33.3.1: (a) The null distribution of the uncorrected F statistic with autoregressive errors, against the nominal $F(5, 115)$ density and the moment-matched $F(\epsilon d_1, \epsilon d_2)$ of Proposition 33.3.3, with the nominal 5% point marked. (b) Simulated level of four tests as ρ grows.

[Run this code online](#)

How much does the failure of sphericity cost? Figure 33.3.1 answers with a simulation under H . Panel (a) shows the null distribution of the uncorrected F : more dispersed than $F(5, 115)$, with a heavier right tail, and tracking Proposition 33.3.3 closely. Panel (b) traces the level of a nominal 5% test as the autoregressive parameter grows. At $\rho = 0$ it is 0.050, as Theorem 33.3.2(b) requires; by $\rho = 0.8$, where $\epsilon = 0.562$, it is 0.085. The Greenhouse–Geisser test is conservative throughout (0.038 to 0.049), the Huynh–Feldt test close to nominal (0.049 to 0.056), and the multivariate test exact at 0.049 for every ρ . A spherical covariance of the form Lemma 33.3.1(iv) with unequal a_j , far from compound symmetric, gives the uncorrected test a level of 0.050: sphericity, not compound symmetry, is what matters.

33.3.5 The modern treatment

The corrections above patch a fixed analysis; the mixed-model approach fits the covariance instead. Write the data as a linear mixed model (Definition 32.2.1) with a fixed effect for each occasion and a within-subject covariance $\mathbf{R}$ of chosen shape, estimate its parameters by restricted maximum likelihood (Definition 32.4.1), and test the fixed effects by a Wald statistic with estimated degrees of freedom (Proposition 32.5.1). Compound symmetric $\mathbf{R}$ reproduces the uncorrected univariate F (Exercise (B2)), unstructured $\mathbf{R}$ the multivariate test up to the degrees-of-freedom approximation, and anything between is a model the classical scheme has no room for. Unbalanced data and missing visits are handled too, whereas Eq. 33.3.1 and T^2 need a complete rectangle. The price is that the covariance must be chosen, and the choice affects the standard errors: Section 33.4.

Remark (Between-subjects factors). With a between-subjects factor the within-subject analysis is the same, applied to the pooled within-group covariance, and the univariate tests are exact iff the pooled Ψ is spherical *and* the groups share a common Σ . Mauchly’s (1940) test of sphericity itself, which most software reports, is sensitive to non-normality and, like any preliminary test, distorts the level of what follows.

33.3.6 Exercises

33.3.6.1 A. Check your understanding

Exercise A1.

Show that every 2×2 covariance matrix satisfies sphericity, so that with $m = 2$ the repeated-measures F test is exact whatever Σ is. Identify the test in that case.

Solution. With $m = 2$, $\Psi = \mathbf{U}^\top \Sigma \mathbf{U}$ is 1×1 , hence a multiple of $\mathbf{I}_1$. By [Theorem 33.3.2\(b\)](#) the test is exact. It is the paired t test: the single contrast is $(y_{i1} - y_{i2})/\sqrt{2}$, and F is the square of the paired t statistic on $N - 1$ degrees of freedom.

Exercise A2.

For $m = 3$ and an autoregressive covariance $\sigma_{jk} = \rho^{|j-k|}$ with $0 < \rho < 1$, compute Ψ for the basis $\mathbf{u}_1 = (1, -1, 0)^\top/\sqrt{2}$, $\mathbf{u}_2 = (1, 1, -2)^\top/\sqrt{6}$ and show that sphericity fails. Which pairwise difference has the largest variance?

33.3.6.2 B. Practice

Exercise B1.

Show directly that compound symmetry $\Sigma = \sigma^2\{(1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top\}$ gives $\Psi = \sigma^2(1 - \rho)\mathbf{I}_{m-1}$, and hence that the split-plot analysis of [Theorem 33.1.2](#) is the repeated-measures F test with $\epsilon = 1$. Which λ of [Lemma 33.3.1](#) does it correspond to?

Solution. $\mathbf{U}^\top \mathbf{1} = \mathbf{0}$, so $\Psi = \mathbf{U}^\top \Sigma \mathbf{U} = \sigma^2(1 - \rho)\mathbf{U}^\top \mathbf{U} = \sigma^2(1 - \rho)\mathbf{I}_{m-1}$. Hence $\epsilon = 1$ and $\lambda = \sigma^2(1 - \rho)$, which is the subplot stratum eigenvalue λ_s of [Lemma 33.1.1](#). In [Lemma 33.3.1\(iv\)](#) take $\mathbf{a} = (\sigma^2\rho/2)\mathbf{1}$.

Exercise B2.

Consider the mixed model $y_{ij} = \mu_j + u_i + e_{ij}$ with u_i and e_{ij} independent normal, of variances σ_u^2 and σ_e^2 . Show that restricted maximum likelihood gives $\hat{\sigma}_e^2 = \text{MS}_E$ and $\hat{\sigma}_u^2 = (\text{MS}_S - \text{MS}_E)/m$ when this is nonnegative, where MS_S is the between-subject mean square, and that the Wald statistic for $H: \mu_1 = \dots = \mu_m$, divided by $m - 1$, is the uncorrected F of [Eq. 33.3.1](#).

33.3.6.3 C. Going deeper

Exercise C1.

For $m = 4$, $N = 20$ and Σ compound symmetric, compare by simulation the power of the uncorrected univariate test, the Greenhouse–Geisser test and the multivariate test of [Proposition 33.3.4](#) against a linear trend in the occasion means. Repeat with an

autoregressive Σ of the same total variance. Which would you choose if you knew the alternative were a linear trend, and why is a single contrast better still?